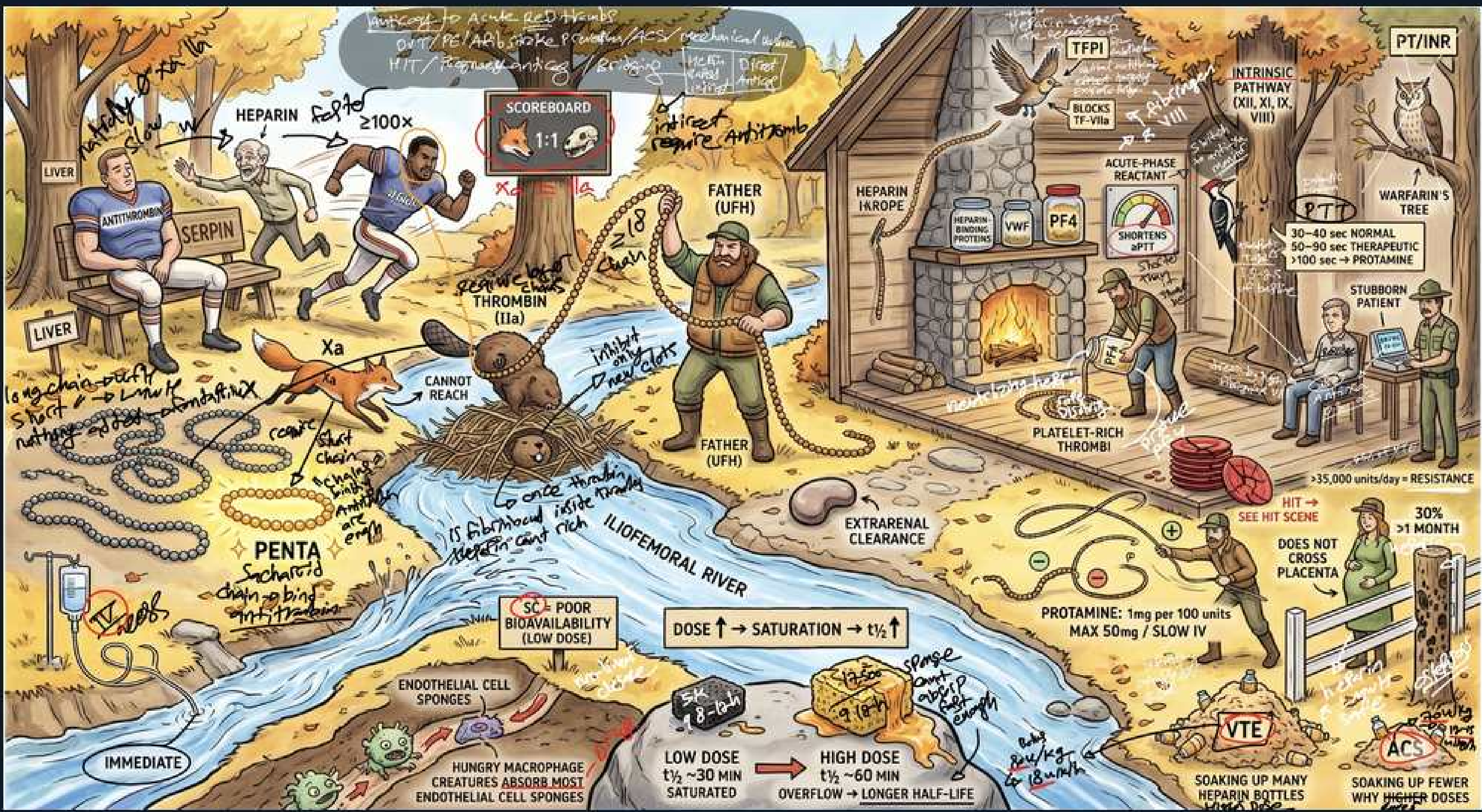
Parenteral — Unfractionated Heparin (UFH) River



1. Antithrombin potentiation ≥100×

WHAT YOU SEE

Two runners 'HEPARIN' + 'SERPIN', '≥100x' banner, '1:1'

WHAT IT ENCODES

Unfractionated HEPARIN works INDIRECTLY: it binds ANTITHROMBIN (a serpin) and accelerates its inhibitory activity by ~1000-fold. UFH is a long chain that can simultaneously bind antithrombin AND thrombin, bridging them so it inactivates BOTH thrombin (IIa) and factor Xa about equally — a 1:1 anti-IIa:anti-Xa ratio.

BOARD TRAP

Heparin has NO intrinsic anticoagulant activity — it requires antithrombin. In antithrombin DEFICIENCY heparin fails (heparin 'resistance'); the fix is antithrombin repletion or a direct thrombin inhibitor, not more heparin.

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2. Inactivates thrombin (IIa) — needs long chain

WHAT YOU SEE

Father figure (UFH) holding chains, 'THROMBIN (IIa)', long chain

WHAT IT ENCODES

To inactivate THROMBIN, heparin must form a ternary complex (antithrombin + heparin + thrombin), which requires a chain of ≥18 saccharide units — only the long UFH chains can do this. This is why UFH inhibits IIa strongly, whereas the short LMWH/fondaparinux chains 'cannot reach' thrombin and act mainly on Xa.

BOARD TRAP

The pentasaccharide ALONE (short chain) inhibits Xa but NOT thrombin — chain length is the discriminator between UFH (anti-IIa + anti-Xa) and LMWH/fondaparinux (predominantly anti-Xa).

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3. Pentasaccharide — the active core

WHAT YOU SEE

'PENTA' sugar chain at lower left, schematic

WHAT IT ENCODES

The minimal antithrombin-binding sequence is the unique PENTASACCHARIDE. Binding it induces the conformational change in antithrombin that drives anti-Xa activity. UFH is a heterogeneous mix of long chains containing this core; only a fraction of UFH chains actually carry the active pentasaccharide.

BOARD TRAP

Anti-Xa inhibition needs only the pentasaccharide; anti-IIa needs the pentasaccharide PLUS a long chain. This single fact explains the whole UFH-vs-LMWH-vs-fondaparinux activity spectrum.

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4. aPTT monitoring 1.5–2.5×

WHAT YOU SEE

Cabin owl 'PT/INR' contrasted, 'aPTT', '30–40s normal / 50–90 therapeutic / >100 protamine'

WHAT IT ENCODES

Therapeutic IV UFH is MONITORED with the aPTT (target ~1.5–2.5× control, roughly 50–90 s) or with an anti-Xa heparin level, adjusted by a weight-based nomogram. Its anticoagulant effect is immediate after IV bolus. Markedly prolonged aPTT (>100 s) flags over-anticoagulation.

BOARD TRAP

UFH is monitored by aPTT (or anti-Xa) — NOT PT/INR (that's warfarin). LMWH/fondaparinux at standard doses need NO routine monitoring; mixing up the assays is a classic error.

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5. Acute-phase reactant → shortens aPTT

WHAT YOU SEE

Cabin panel 'ACUTE-PHASE REACTANT', 'SHORTENS aPTT', 'PF4'

WHAT IT ENCODES

Heparin binds many plasma proteins, several of which are ACUTE-PHASE REACTANTS (e.g., factor VIII, fibrinogen, platelet factor 4). In acute illness these rise and bind/neutralize heparin, blunting and SHORTENING the aPTT response — a mechanism of 'heparin resistance' requiring higher doses or anti-Xa-guided dosing.

BOARD TRAP

An unexpectedly high heparin requirement (short aPTT despite big doses) suggests acute-phase protein binding or antithrombin deficiency — switch to anti-Xa monitoring rather than blindly escalating.

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6. Protamine — full reversal

WHAT YOU SEE

'PROTAMINE: 1mg per 100 units', 'MAX 50mg / SLOW IV'

WHAT IT ENCODES

PROTAMINE SULFATE is the antidote for UFH: ~1 mg neutralizes ~100 units of heparin (dose by heparin given in the last 2–3 h), max ~50 mg, given SLOWLY IV. It is a positively charged protein that binds and inactivates the negatively charged heparin, reversing it COMPLETELY and rapidly.

BOARD TRAP

Protamine FULLY reverses UFH but only PARTIALLY reverses LMWH (~60%) and does NOT reverse fondaparinux at all. Rapid infusion causes hypotension/anaphylaxis; fish-allergy/NPH-insulin/prior-vasectomy patients are higher risk.

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7. Short half-life / fast offset

WHAT YOU SEE

'cannot reach', short-chain note, river current

WHAT IT ENCODES

IV UFH has a SHORT, dose-dependent half-life (~60–90 min) with rapid on/off, so it is ideal when anticoagulation may need to be stopped quickly (e.g., periprocedurally, high bleeding risk, renal failure). The drip is simply turned off to reverse most effect within a few hours.

BOARD TRAP

Because of its short half-life and protamine reversibility, UFH (not LMWH/DOAC) is preferred when the patient has SEVERE RENAL FAILURE or imminent procedure/high bleeding risk — quick offset is the deciding feature.

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8. Extra-renal clearance

WHAT YOU SEE

'EXTRARENAL CLEARANCE' label by the cabin

WHAT IT ENCODES

UFH is cleared mainly by the RETICULOENDOTHELIAL system and liver (extra-renal), with renal clearance only at high doses. This non-renal clearance is exactly why UFH is the preferred parenteral anticoagulant in severe RENAL IMPAIRMENT, where LMWH and fondaparinux accumulate.

BOARD TRAP

In CrCl <30, choose UFH (extra-renal clearance, protamine-reversible) over LMWH/fondaparinux (renally cleared, accumulate). This renal point is heavily tested.

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9. Heparin-induced thrombocytopenia risk

WHAT YOU SEE

'HIT — SEE HIT SCENE', '>35,000 units/day → RESISTANCE', PF4

WHAT IT ENCODES

UFH carries the HIGHEST risk among the parenteral anticoagulants for the immune heparin-induced thrombocytopenia reaction: IgG antibodies against the heparin–PF4 complex activate platelets, causing a paradoxical PROTHROMBOTIC state with a platelet drop (typically a >50% fall, days 5–10). Stop ALL heparin and switch to a non-heparin agent (argatroban/bivalirudin/fondaparinux).

BOARD TRAP

In suspected heparin-induced thrombocytopenia, never give platelet transfusions or warfarin first (warfarin alone can cause limb gangrene) and never 'just switch to LMWH' — LMWH cross-reacts. Use a direct thrombin inhibitor. Note >35,000 units/day signals heparin RESISTANCE (often antithrombin deficiency).

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10. Does NOT cross placenta — safe in pregnancy

WHAT YOU SEE

Cabin sign 'DOES NOT CROSS PLACENTA'

WHAT IT ENCODES

Heparins (UFH and LMWH) are large, negatively charged molecules that DO NOT cross the placenta, making them the anticoagulants of choice in PREGNANCY. LMWH is generally preferred over UFH antepartum for convenience and lower osteoporosis/heparin-induced-thrombocytopenia risk; UFH is favored near delivery for its short, reversible action.

BOARD TRAP

Pregnancy → heparin (LMWH preferred), NOT warfarin (teratogenic) and NOT DOACs (insufficient data, placental crossing). This is the standard pregnancy-anticoagulation pivot.

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11. Osteoporosis with long-term use

WHAT YOU SEE

'30% >1 MONTH' osteoporosis note on cabin wall

WHAT IT ENCODES

Prolonged UFH therapy (□1 month) can cause OSTEOPOROSIS and vertebral fractures (heparin disrupts osteoblast/osteoclast balance). This adverse effect is more pronounced with UFH than with LMWH, and is a reason to prefer LMWH for long-term parenteral anticoagulation (e.g., in pregnancy).

BOARD TRAP

Long-term heparin osteoporosis is greater with UFH than LMWH — another reason LMWH is preferred for extended courses such as pregnancy prophylaxis.

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12. SC poor bioavailability (low dose)

WHAT YOU SEE

Bottom 'SC = POOR BIOAVAILABILITY (LOW DOSE)', sponging macrophages

WHAT IT ENCODES

Given SUBCUTANEOUSLY at low (prophylactic) doses, UFH has POOR, saturable bioavailability because endothelial cells and macrophages bind and clear it (the 'sponging' cells). This saturable, dose-dependent clearance explains its NONLINEAR kinetics — half-life lengthens as dose rises.

BOARD TRAP

UFH kinetics are NON-linear/saturable (zero-order at higher doses), unlike LMWH's predictable linear kinetics — this is why UFH needs aPTT monitoring and LMWH usually does not.

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13. Dose ↑ → saturation → longer half-life

WHAT YOU SEE

'DOSE↑ → SATURATION', 'LOW DOSE → HIGH DOSE → OVERFLOW → LONGER HALF-LIFE'

WHAT IT ENCODES

As the UFH dose increases, the rapid saturable clearance mechanism gets overwhelmed ('overflow'), shifting clearance to the slower renal route and PROLONGING the half-life. This dose-dependent (nonlinear) clearance is why therapeutic UFH must be titrated to aPTT/anti-Xa with weight-based nomograms.

BOARD TRAP

Don't assume a fixed half-life for UFH — higher doses disproportionately prolong it. This nonlinearity, plus protein binding, mandates lab-guided titration that LMWH/fondaparinux don't need.
